AV

AutoValuate
Intelligence

Know what your car is really worth — with the reasoning shown, not hidden.

A hybrid computer-vision + explainable-ML + agentic-RAG valuation platform for the UAE. · Live product.

SP Jain School of Global Management · Group Project

THE TEAM

Four builders, one system

KM

Krishna Mathur

AS25DXB018

Deep learning — the on-device damage detector (YOLOv8)

YP

Yash Petkar

AS25DXB020

Valuation model, data pipeline & the live product build

AS

Atharva Soundankar

AS25DXB021

Agentic backend, orchestration & the RAG retrieval layer

?

[Fourth member]

AS25DXB0__

Frontend, UX & product — to be credited

All four of us touched every layer — these were our anchors.

THE PROBLEM

A used-car seller has a feeling. The dealer has the data.

- The UAE used-car market turns over ~1.5 million cars a year — and pricing is opaque.
- Sellers negotiate on a hunch; dealers do it all day and win every time.
- Existing tools give one number with no reasoning, no damage awareness, and no honesty about uncertainty.
- "How much is my car worth, and can I defend that number?" has no trustworthy answer.

The gap

Nobody gives the seller an explainable, damage-aware price they can actually argue with — for free.

~1.5M

used cars sold / year (UAE)

AED 8k

typical mispricing per car

0

tools that show their working

100%

free to the seller

THE PRODUCT · LIVE

Photos in. A defensible price out.

- Snap your car; a trained detector scans it for damage on your device.
- An explainable model prices it, showing every factor that moved the number.
- Live comparable listings and a written report — every figure traceable.
- Runs on free infrastructure; the whole thing is live today.

EXPLAINABLE PRICING

Not a number — a number with its reasons

Every driver, quantified

SHAP shows exactly how each feature — cylinders, age, model, mileage — pushed the price up or down, in dirhams.

Honest uncertainty

A calibrated 78% confidence range from split-conformal prediction — not a false-precision point estimate. Median error is stated on the card.

DEEP LEARNING · RUNS IN YOUR BROWSER

A damage detector that never sees your photos

- YOLOv8-small, fine-tuned on ~18k images (CarDD + VehiDE), 8 damage classes.
- mAP@0.5 = 0.732 — a real, honestly-reported number.
- Exported to ONNX and run with onnxruntime-web — inference happens in the browser, so photos never leave the device.
- This also makes computer vision free at any scale: no server GPU, no per-image cost.
- The detected condition feeds straight into the price.

FROM DETECTION TO DECISION

What the damage costs — and whether to fix it

REPAIR ESTIMATE

What the detected damage would cost to put right

3 items

Estimated repair cost

AED 2,300 – AED 10,160

Dent ×3	severe	AED 1,440 – AED 5,760
Lamp Broken	severe	AED 560 – AED 3,200
Scratch ×2	moderate	AED 300 – AED 1,200

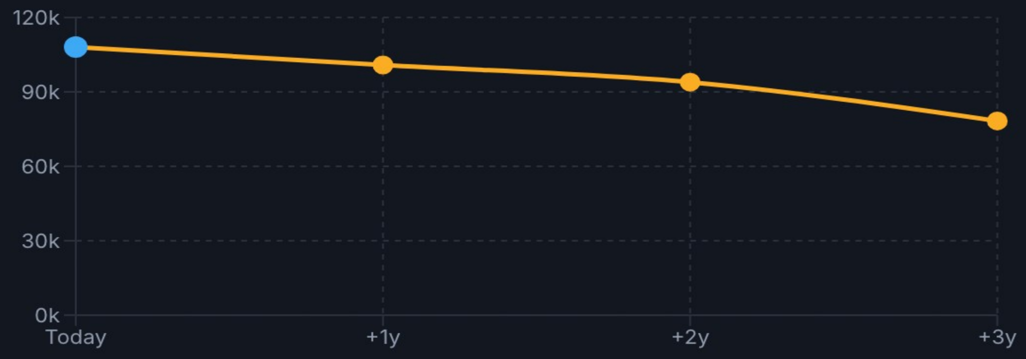
This damage is costing you about AED 8,763 in value, against a repair bill of roughly AED 6,230. Repairing before you sell is likely to pay for itself.

Indicative UAE independent-workshop prices from a published table, scaled by detected severity and instance count — not a quote. Get a workshop to confirm before you decide.

SELL-TIMING FORECAST

What the pricing model says this car will be worth as it ages

−7% / year



Holding this car costs roughly AED 602 a month in lost value (about AED 7,220 over the next year). It's holding value reasonably well, so there's less urgency to sell immediately.

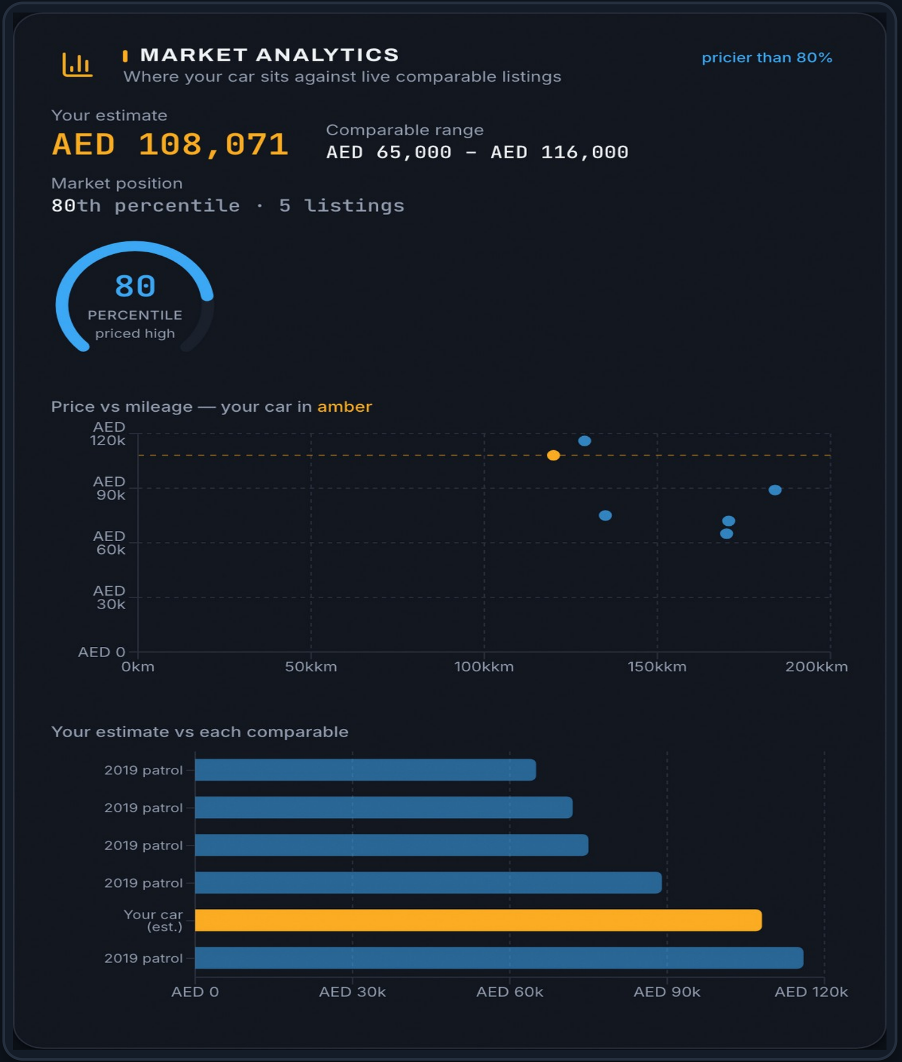
Projected by re-running the pricing model with this car aged forward (age +N years, mileage +N × its own annual rate) — not a generic depreciation rate. It carries the same 19.57% median error as any other estimate, and it assumes the market itself stays flat.

Repair estimate: detection → itemised AED cost → a worth-fixing verdict.

Forecast: the same model, this car aged forward — real depreciation, not a rule of thumb.

MARKET CONTEXT

Where your car sits in the real market



Price vs mileage

Your car plotted against live Dubizzle comparables, inside the model's fair-value band.

Market position

A percentile gauge: is this priced high or low versus genuinely comparable cars?

Estimate vs each comp

Your valuation bar-charted against every retrieved comparable listing.

All theme-aware

Every chart is responsive and renders in both light and dark mode.

An AI that cannot invent a number

SELLER REPORT

Structured from computed evidence · every figure grounded

Certificate | PDF | Share | verified

Based on the details provided, the 2019 Nissan Patrol has an estimated fair-market value between AED 69,433 and AED 168,209, with a mid-point of AED 108,071. This range is a calibrated 78% confidence interval.

The main factors behind this estimate are NoOfCylinders (+122,451 AED), Age (-20,323 AED), Kilometers (-16,888 AED). On held-out testing the pricing model carries a median error of about 19.57%, so treat the mid-point as a guide, not a guarantee.

The photo assessment gave a condition score of 93/100, reflecting detected damage that affects value.

Comparable live listings support this range: 2019 Nissan Patrol 129,000 km at AED 116,000; 2019 Nissan Patrol 184,000 km at AED 89,000; 2019 Nissan Patrol 135,000 km at AED 75,000; 2019 Nissan Patrol 170,596 km at AED 72,000; 2019 Nissan Patrol 170,000 km at AED 65,000. If confidence is limited or the car has damage beyond what the photos show, a professional inspection is the safest next step before you set a final asking price.

13 numbers and 14 citations checked by the Verifier — all trace to a computed value.

ASK THE ASSISTANT

Grounded in this valuation — every number is checked by the Verifier

grounded

Is this a good deal?

Against this car's computed range, the fair mid-point is AED 108,071 V2 , with a calibrated band from AED 69,433 V1 to AED 168,209 V3 . Live comparables: 2019 nissan patrol 129000km at AED 116,000 C1 ; 2019 nissan patrol 184000km at AED 89,000 C2 ; 2019 nissan patrol 135000km at AED 75,000 C3 . Anything meaningfully above AED 168,209 V3 is above the model's fair band — use the mid-point as your anchor and the upper bound as your walk-away.

7 numbers · 7 citations verified · via template

Ask about the price, damage, or comparables...

The Verifier gate

Every number in the report and the chat is checked against computed evidence. An ungrounded figure is rejected before you ever see it — faithfulness 1.000.

UNDER THE HOOD

Three models, one honest pipeline

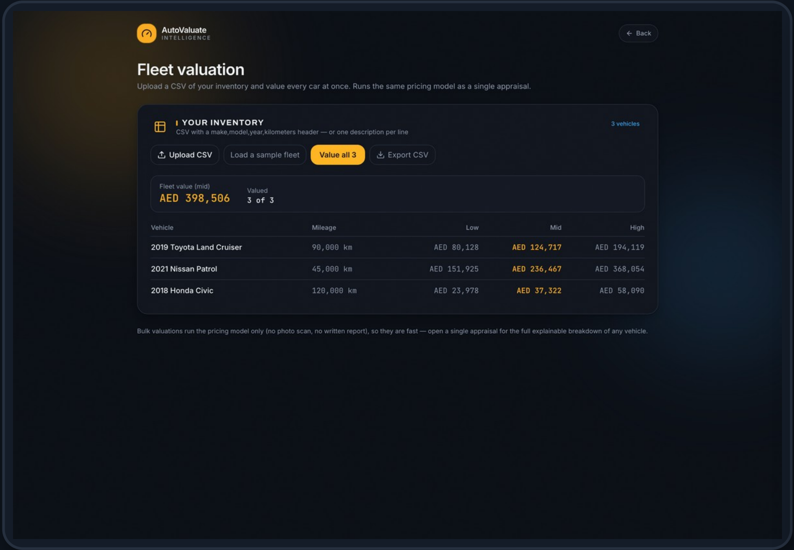
1	→	2	→	3	→	4	→	5
Intake		**Vision (CV)**		**Pricing (ML)**		**Retrieval (RAG)**		**Report + Verifier**
validate & normalise the vehicle		YOLOv8 → ONNX, on-device		XGBoost quantile + conformal + SHAP		hybrid: embeddings + BM25 + structured		LLM grounded, deterministically checked

Deep-learning & ML applied: CNN object detection · transfer learning · IoU / NMS · mAP · ONNX quantization · gradient-boosted trees · quantile regression · split-conformal prediction · SHAP attribution · sentence embeddings · BM25 · cross-encoder reranking · LangGraph agents · retrieval-augmented generation · deterministic verification.

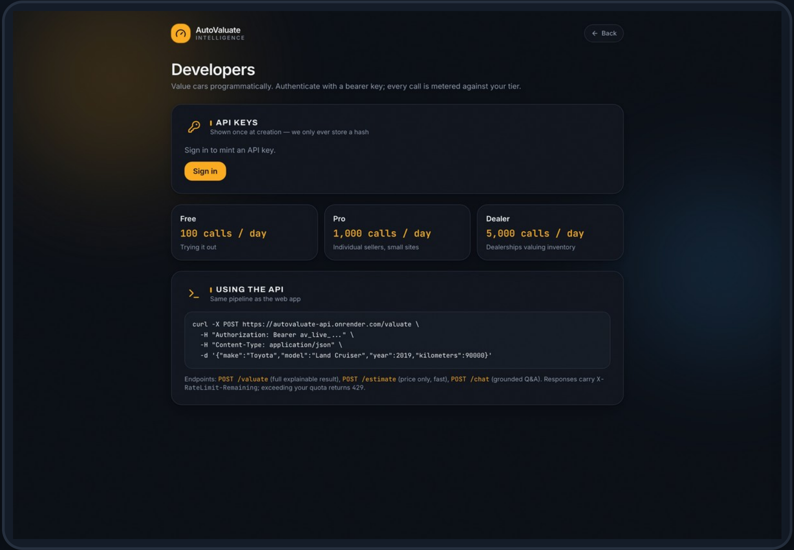
Stack: Next.js (Vercel) · FastAPI (Render) · Supabase · Kaggle GPU · GitHub Actions — all free tier.

BUSINESS MODEL · SAAS

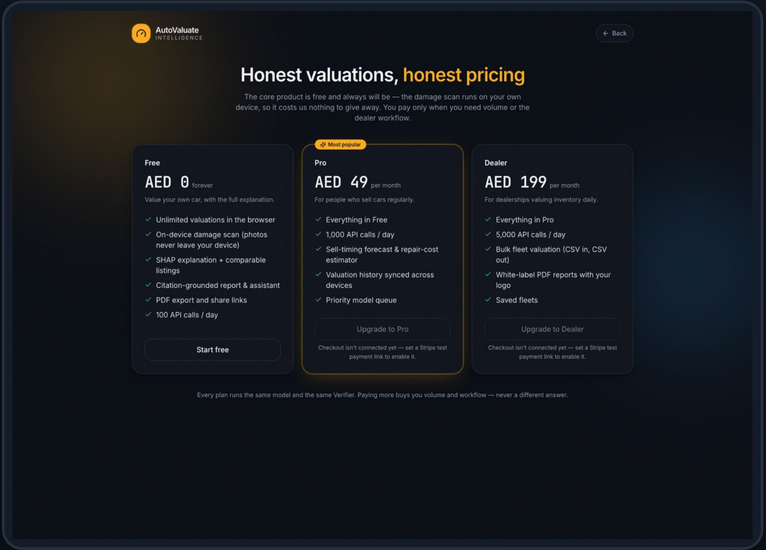
Free for sellers. Paid for volume.



Dealer fleet valuation



API keys + usage metering



Plans: Free / Pro / Dealer

The wedge

The on-device CV makes the core product free to run — so we give it away and monetise the workflows businesses actually pay for: bulk fleet valuation, a metered API, and white-label reports. Every tier runs the same model and the same Verifier — paying more buys volume, never a different answer.

WE MEASURED, AND REPORTED WHAT WE FOUND

Honesty as a feature

0.732	1.000	80.0%	0
CV mAP@0.5 — reported, not rounded up	report faithfulness (Verifier)	conformal coverage (target 80%)	WCAG 2.1 AA violations

What the research found (D3)

Raw quantile regression — the obvious thing to ship — promises 80% coverage and delivers 54.8%. The "±25% rule of thumb" delivers 56.3% and looks tight precisely because it's wrong. Split-conformal hits 80.0%. Without calibration the product would be confidently wrong in a way no user could detect.

What the research found (D5)

We proved the retriever is at its mathematical ceiling: the benchmark's max score is 0.780 (porsche has 0 listings, 23 of 37 makes have <5), and it scores exactly 0.780. Retrieval is data-bound, not algorithm-bound — corpus growth is the only lever. Both findings argued against the obvious design choice.

WHERE IT STANDS

A real, honest, working MVP

90

/ 100

as a capstone product

Real MVP? ✓ Yes

Live, usable end-to-end today: a stranger can value a car, see the reasoning, and get a defensible number. Not a mockup — a running system on real data.

SaaS-ready? ~72 / 100

Auth, API keys, metering, plans, dealer bulk and white-label all built. Remaining: real payments, a bigger corpus, and production CV scale.

- Everything shown is live, free-tier, and reproducible — the metrics page is public.
- The honest gaps are named, not hidden: thin corpus (Phase E cron is growing it), test-mode payments, single-region.

Thank you. · AutoValuate Intelligence · auto-valuate-intelligence.vercel.app